

Healthcare Data Analytics Assessment Report

Submitted By

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Role: Data Analytics Internship Assessment

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Tools Used: Python, Pandas, NumPy, Matplotlib, Jupyter Notebook, Power BI

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1. Introduction

Healthcare organizations generate large volumes of patient and treatment data every day. Analyzing this data helps hospitals improve operational efficiency, understand treatment costs, identify disease trends, and make better business decisions.

This project focuses on analyzing a healthcare dataset using Python and Power BI. The dataset was cleaned, explored, and visualized to answer business and demographic questions provided in the assessment.

2. Project Objectives

The objectives of this project are:

- Clean and preprocess the healthcare dataset.
- Perform exploratory data analysis.

- Identify treatment cost trends.
 - Analyze department-wise revenue.
 - Compare treatment costs across diseases, cities, genders, and insurance status.
 - Build an interactive Power BI dashboard.
 - Generate meaningful business insights.
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3. Dataset Description

The dataset contains patient-level healthcare records including:

- Patient ID
- Age
- Gender
- Disease
- Department
- City
- Treatment Cost
- Insurance Status
- Blood Group
- Outcome
- Admission Date
- Discharge Date

The dataset consists of healthcare records that are suitable for business and demographic analysis.

(Insert a screenshot of the first few rows of the dataset.)

4. Technology Stack

The following tools and technologies were used:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Power BI
- Git & GitHub

Python was used for data cleaning, preprocessing, and analysis, while Power BI was used to build an interactive dashboard.

5. Data Cleaning & Preprocessing

The dataset was cleaned before performing any analysis.

The following steps were completed:

- Imported required libraries.
- Loaded the dataset.
- Inspected the dataset using `.info()`, `.describe()`, and `.head()`.
- Checked missing values.
- Removed duplicate records.
- Corrected data types.
- Converted date columns.
- Standardized categorical values.
- Created the **Age_Group** column.
- Validated the cleaned dataset.
- Exported the cleaned dataset for Power BI.

(Insert screenshots of cleaning steps from the notebook.)

6. Exploratory Data Analysis (EDA)

EDA was performed to understand the characteristics of the dataset.

The following analyses were completed:

Univariate Analysis

- Disease distribution
- Gender distribution
- Department distribution
- Blood group distribution
- Insurance status distribution

Bivariate Analysis

- Disease vs Treatment Cost
- Department vs Revenue
- Gender vs Treatment Cost
- Age Group vs Healthcare Expenditure

- City vs Average Treatment Cost

(Insert Python charts here.)

7. Business Analysis

Q1. Highest Treatment Cost by Disease

Finding: Asthma contributes the highest total treatment cost (**10,938,576.50**).

Insight: Asthma accounts for the largest share of healthcare expenditure in the dataset.

Q2. Highest Revenue Department

Finding: Neurology generates the highest revenue (**11,955,993.50**).

Insight: Neurology contributes the highest revenue among all hospital departments.

Q3. Average Treatment Cost by Disease and Insurance Status

Treatment costs vary across diseases and insurance status. Migraine has the highest average treatment cost among insured patients, while Hypertension has the highest average treatment cost among uninsured patients.

Q4. Insurance Coverage

Insured patients cover **51.87%** of the total treatment cost.

This indicates that insurance plays a significant role in financing healthcare expenses.

Q5. Highest Average Treatment Cost by City

Delhi records the highest average treatment cost per patient (**138,487.95**).

8. Patient Demographics Analysis

Q6. Healthcare Expenditure by Age Group

Patients aged **60+** contribute the highest healthcare expenditure.

Q7. Disease Distribution Across Age Groups

Asthma is the most common disease across most age groups, while Arthritis is most common among patients aged 46–60.

Q8. Treatment Cost by Gender

Female patients have a slightly higher average treatment cost than male patients. However, the difference is relatively small.

Q9. Critical Outcomes by Blood Group

AB– blood group records the highest number of critical outcomes in the dataset.

9. Power BI Dashboard

An interactive dashboard was developed containing:

- KPI Cards
- Total Treatment Cost by Disease
- Revenue by Department
- Average Treatment Cost by City
- Healthcare Expenditure by Age Group
- Average Treatment Cost by Gender
- Critical Outcomes by Blood Group
- Disease Distribution by Age Group
- Insurance Status Distribution
- Interactive Slicers

(Insert a full-page screenshot of your dashboard.)

10. Key Insights

- Asthma has the highest treatment cost.
 - Neurology generates the highest revenue.
 - Insured patients account for 51.87% of treatment costs.
 - Delhi has the highest average treatment cost.
 - The 60+ age group contributes the highest healthcare expenditure.
 - Asthma is the most common disease across most age groups.
 - Female patients have a slightly higher treatment cost than male patients.
 - AB– blood group records the highest critical outcomes.
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11. Challenges Faced

- Handling missing values and duplicates.
- Standardizing categorical data.

- Creating appropriate age groups.
 - Building an interactive Power BI dashboard.
 - Selecting the most suitable visualizations for business insights.
-

12. Conclusion

The healthcare dataset was successfully cleaned, analyzed, and visualized using Python and Power BI. The project identified key trends related to treatment costs, department revenue, insurance coverage, patient demographics, and disease distribution. The Power BI dashboard provides an interactive way to explore the data and supports informed decision-making.

13. Future Improvements

Future enhancements may include:

- Predictive models for treatment cost estimation.
- Patient risk prediction.
- Readmission prediction.
- Real-time dashboard integration.
- SQL database connectivity.
- Advanced Power BI dashboards with drill-through and forecasting features.

Healthcare Data Analytics Dashboard

500

Total Patients

61.23M

Total Treatment Cost

61.23M

Total Revenue

122.46K

Average Treatment...

0.52

Insurance Covera...

City

☐ Bengaluru
☐ Delhi
☐ Hyderabad
☐ Kolkata
☐ Mumbai
☐ Patna
☐ Pune

Disease

☐ Arthritis
☐ Asthma
☐ Covid-19
☐ Diabetes
☐ Heart Disease
☐ Hypertension
☐ Migraine

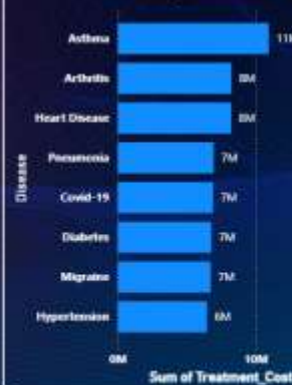
Department

☐ Cardiology
☐ General Medic...
☐ Neurology
☐ Oncology
☐ Orthopedics

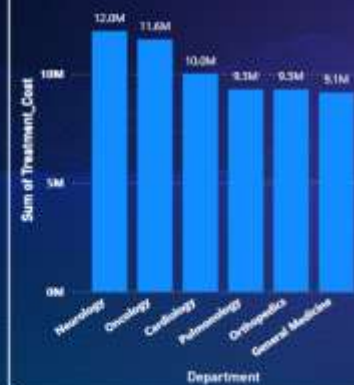
Insurance_Status

☐ No
☐ Yes

Total Treatment Cost by Disease



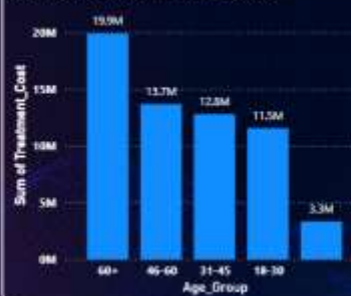
Revenue by Department



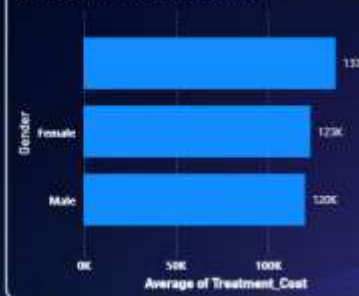
Average Treatment Cost by City



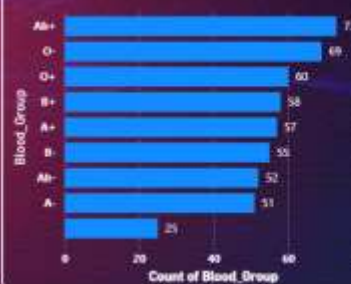
Healthcare Expenditure by Age Group



Average Treatment Cost by Gender



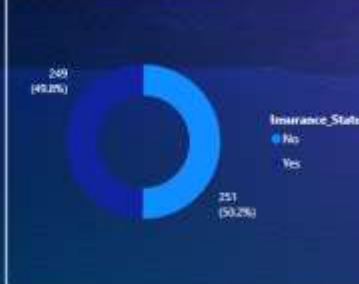
Critical Outcomes by Blood Group



Patients by Insurance Status



Patients by Insurance Status



Key Insights

- Asthma has the highest treatment cost.
- Neurology generates the highest revenue.
- Delhi has the highest average treatment cost.
- Patients aged 60+ contribute the highest healthcare expenditure.